

CODEEVOLVE: an open-source evolutionary framework for algorithmic discovery and optimization

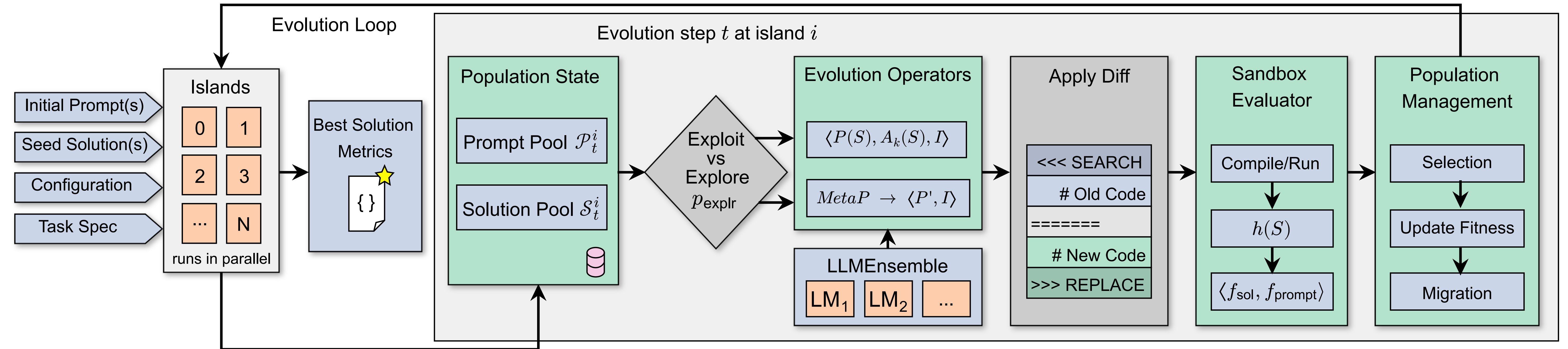
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Evolve executable programs with LLM-guided search



Depth exploitation

We apply rank-based sampling to select a parent solution, then use it together with a number of ancestors to generate a new child. The ancestral context encourages targeted, incremental improvements as depth increases.

Meta-prompting exploration

A parent prompt is randomly sampled and enriched by an auxiliary LLM. The new prompt is combined with a randomly selected parent solution to generate a new child. We exclude the ancestor chain to allow exploration of novel strategies while leveraging a richer prompt.

Inspiration-based crossover

We provide the ensemble with inspiration solutions drawn either by rank or uniformly at random. This achieves the semantic intent of crossover by allowing the ensemble to incorporate successful patterns from other solutions through targeted edits to the parent.

Population and model orchestration

Islands evolve independently in parallel and exchange strong solutions according to a migration topology. CVT-MAP-Elites partitions feature space using centroidal Voronoi tessellations, efficiently preserving diverse elites. Weighted LLM ensembles can be configured separately for exploration and exploitation, with an auxiliary LLM for meta-prompting.

1 Competitive with reported discovery systems

Problem	AlphaEvolve	ThetaEvolve	CODEEVOLVE	
			Qwen3-Coder-30B	Gemini-2.5
CirclePackingSquare ($n = 26$) $\uparrow$	2.63586	2.63598	2.63598	2.63597
CirclePackingSquare ($n = 32$) $\uparrow$	2.93794	—	2.93956	2.93950
CirclePackingRect ($n = 21$) $\uparrow$	2.36583	—	2.36339	2.36583
HexagonPacking ($n = 11$) $\downarrow$	3.93009	—	4.02786	3.93794
HexagonPacking ($n = 12$) $\downarrow$	3.94191	—	4.01092	4.00001
MinimizeMaxMinDist ($n = 16, d = 2$) $\downarrow$	12.88927	—	12.88925	12.88923
MinimizeMaxMinDist ($n = 14, d = 3$) $\downarrow$	4.16585	—	4.16765	4.16579
FirstAutocorrIneq $\downarrow$	1.50316	1.50313	1.51343	1.55438
SecondAutocorrIneq $\uparrow$	0.96102	0.94690	0.88110	0.87067

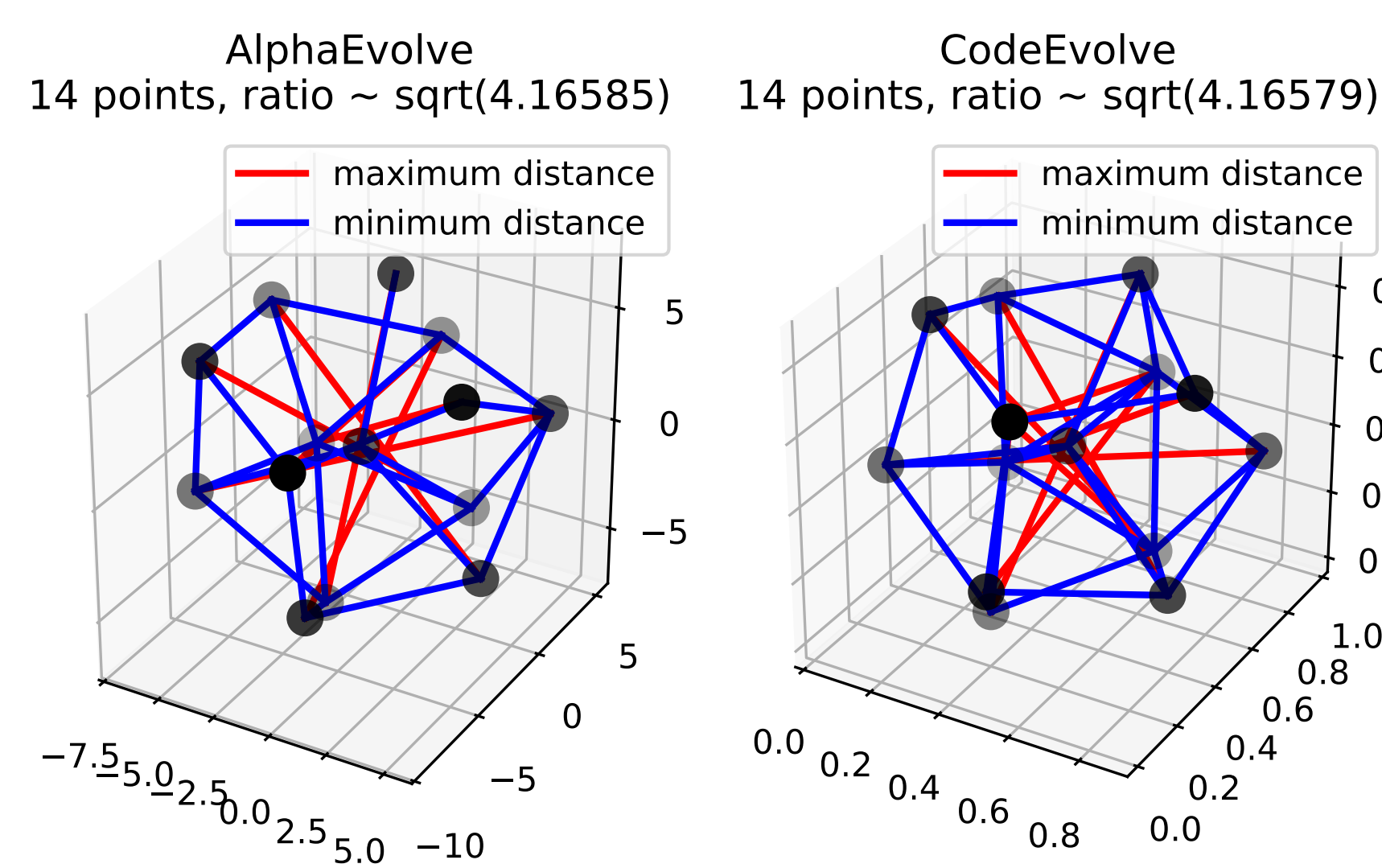
Best solutions; bold indicates the best value in each row. $\uparrow$: higher is better; $\downarrow$: lower is better. AlphaEvolve and ThetaEvolve use reported results, with different models and budgets.

2 Strong results under matched conditions

Problem	ShinkaEvolve		OpenEvolve		CODEEVOLVE	
	Mean	Best	Mean	Best	Mean	Best
CirclePackingSquare ($n = 26$) $\uparrow$	2.47568	2.61551	2.54962	2.62450	2.63576	2.63598
CirclePackingSquare ($n = 32$) $\uparrow$	2.53205	2.69473	2.84897	2.93156	2.93751	2.93957
CirclePackingRect ($n = 21$) $\uparrow$	2.21694	2.32701	2.28131	2.36513	2.36212	2.36465
HexagonPacking ($n = 11$) $\downarrow$	5.55234	5.00000	4.30696	4.20679	4.02491	4.01377
HexagonPacking ($n = 12$) $\downarrow$	5.07507	4.38608	4.01991	4.00010	4.10841	4.05198
MinimizeMaxMinDist ($n = 16, d = 2$) $\downarrow$	13.48234	13.11627	12.97046	12.94042	12.91569	12.90274
MinimizeMaxMinDist ($n = 14, d = 3$) $\downarrow$	4.37160	4.21087	4.18983	4.17140	4.26257	4.16765
FirstAutocorrIneq $\downarrow$	1.52173	1.51556	1.58031	1.57510	1.54696	1.51642
SecondAutocorrIneq $\uparrow$	0.82010	0.84646	0.84634	0.86086	0.86533	0.90114

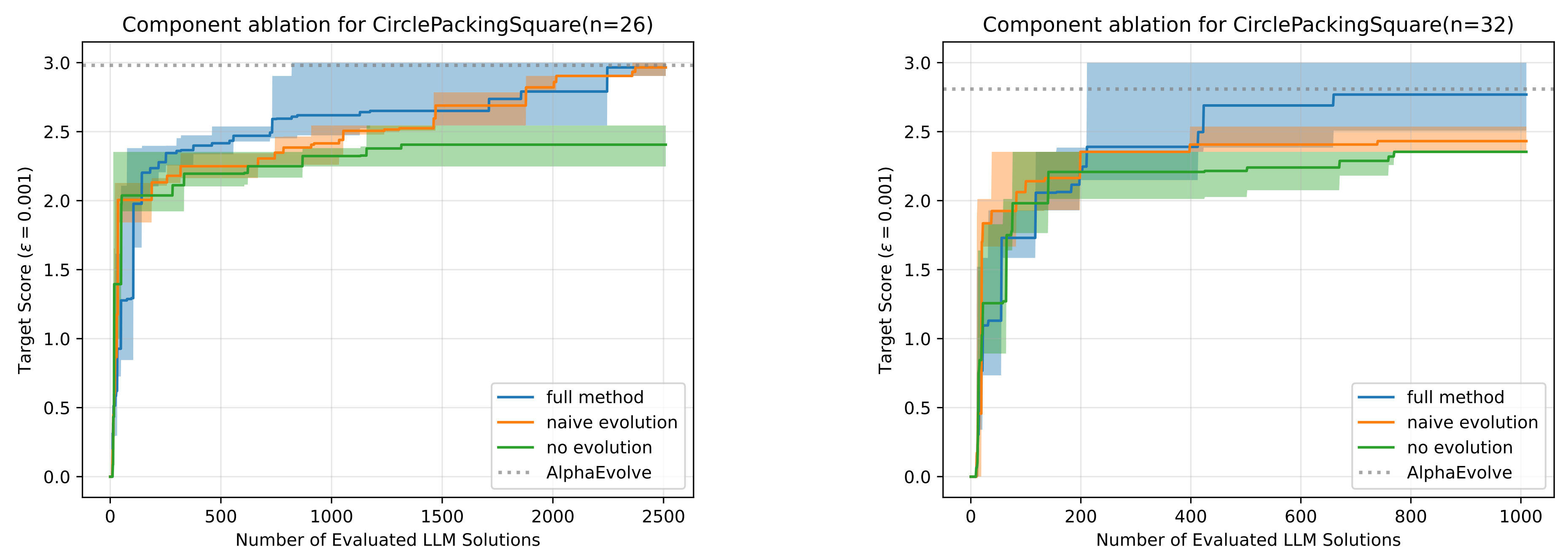
Qwen3-Coder-30B for all frameworks; matched configurations and evaluation budgets; three independent runs. Bold indicates the best mean and best solution separately.

Discovered configurations



MinimizeMaxMinDist ($n = 14, d = 3$). Best solutions from CODEEVOLVE and AlphaEvolve; lower is better.

Component ablations



Combining components improves search. Qwen3-Coder-30B; mean across three runs, with shading showing the best and worst results.

Evolutionary search helps smaller models rival frontier systems

Competitive discovery with open weights. Within CODEEVOLVE, Qwen3-Coder-30B outperforms the Gemini-2.5 ensemble on **4 of 9 instances**. It also exceeds reported AlphaEvolve results on **both CirclePackingSquare instances and one MinimizeMaxMinDist instance**, showing that a smaller open-weight backbone can support frontier-level results on selected discovery tasks.

How we search matters. With the same Qwen backbone and matched budgets, CODEEVOLVE leads OpenEvolve and ShinkaEvolve in best solution and mean on **6 of 9 instances**. Component ablations support combining exploration, refinement, and inspiration—a practical route to stronger discovery without relying solely on larger models.



Code & data